

Problem 1 (§3.9, #1 (adapted)). Show that if $n + 1$ integers are chosen at random, then at least two of them will have the same remainder when divided by n .

Problem 2 (§3.9, #3 (adapted)). What is the fewest number of times you must roll a twenty-sided dice before you can be assured that 10 or more of the rolls resulted in the same number?

Problem 3 (§3.9, #5 (adapted)). Eleven numbers are chosen from the set $\{1, 2, 3, \dots, 20\}$.

- a) Prove that some two of the eleven numbers add to 21.
- b) Prove that some two of the eleven numbers have a difference of 10.